Cambridge (CIE) A Level Computer Science

Your notes

Database Concepts

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Limitations of file-based systems

What is a file-based system?

- A file-based system refers to any system where **data is stored in separate files** (e.g. text files, spreadsheets)
- Each application or department may have its **own files**, often with **redundant or duplicated data**
- There is **no central control** or shared structure, files are often standalone
- Common in older systems or small-scale setups

Limitations

Limitation	Explanation
Data redundancy	The same data is stored in multiple files, leading to duplication
Data inconsistency	If one file is updated but others aren't, the data becomes unreliable
Lack of data integrity	It's harder to ensure data is accurate and consistent across all files
Poor security	Limited control over who can access or modify different files
Difficult to update	Changes must be made in several places, increasing time and risk of error
Limited scalability	Not suitable for large volumes of data or complex data relationships
No central control	Each application manages its own data, leading to lack of coordination
Hard to manage relationships	Cannot easily link related data across files (e.g. customers and their orders)

Flat file vs Relational

- A **flat file database** is one that stores all data in a **single table**

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- It is simple and easy to understand but causes data redundancy, inefficient storage and is harder to maintain
- A **relational database** is one that organises data into **multiple tables**
  - It uses keys to connect related data which reduces data **redundancy**, makes efficient use of storage and is easier to maintain
- Consider this example **flat file** table of students

ID	first_name	last_name	Personal tutor	FormRoom
1	sam	smith	Roger Hinds	6b
2	fred	lynch	Jess Little	8j
3	depak	noor	Roger Hinds	6b
4	archie	henns	Mary Kent	8k
5	helga	jordan	Mary Kent	8k
6	lizzy	bell	Mary Kent	8k
7	xavier	horten	Jack Berry	3m

- This table has **redundant data** – the tutor and form room information repeats
- This is **inefficient**
- If a tutor changed their name we would need to find all instances of that name and change them all
- Missing any would mean the table had **inconsistent data**
- A **relational** database would solve this issue:
  - A new **table** could be created to store the tutor information and the tutor information in the student table could be moved to the new **table**
  - A **foreign key** in the student **table** (TutorID) could link a student to their tutor

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ID	first_name	last_name	TutorID
1	sam	smith	1
2	fred	lynch	2
3	depak	noor	1
4	archie	henns	3
5	helga	jordan	3
6	lizzy	bell	3
7	xavier	horten	4

TutorID	Tutor Name	FormRoom
1	Roger Hinds	6b
2	Jess Little	8j
3	Mary Kent	8k
4	Jack Berry	3m

- Now the name of each tutor and their form room is stored **only once**
- This means if they change only one piece of data, the data is **updated in the entire database** and **inconsistency** is avoided

Database terminology

Term	Definition
Entity	A real-world object or concept that data is stored about (e.g. Student, Book)
Table	A collection of data about an entity, organised in rows and columns
Record (Tuple)	A single row in a table representing one instance of an entity
Field (Attribute)	A single column in a table, storing one piece of data about the entity

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Primary key	A unique identifier for each record in a table (e.g. StudentID)
Candidate key	A field (or combination of fields) that could be used as a primary key
Secondary key	A field used for searching or sorting, but not necessarily unique
Foreign key	A field that links to the primary key in another table to create relationships
Relationship	A logical connection between two tables/entities
One-to-One	One record in a table relates to one record in another table
One-to-Many	One record in a table relates to many records in another table
Many-to-Many	Records in one table relate to many records in another, and vice versa
Referential integrity	Ensures foreign keys match a primary key in the related table to prevent broken links
Indexing	A technique to speed up searching in a table by creating an ordered list of key fields

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Entity-relationship (E-R) diagrams

What is an entity?

- An entity in something worthy of **capturing and storing data about** e.g. *students, orders, products, courses, customers*
- Entities become **tables** in a **relational database**
- Relational databases store different entities in **separate tables**
- Linking tables depends on the **relationships between entities**
- There are 3 types of (sometimes called degrees of) **relationships:**
  - **One-to-one**
  - **One-to-many**
  - **Many-to-many**
- Imagine a company has
  - A table of products
  - A table of customers
  - A table of the orders the customers have made
- What is the relationship between a **customer** and an **order**?
  - **One** customer can make multiple (**many**) orders
  - But each order relates to a specific (**one**) customer
  - So the relationship between customer and order is **one-to-many**
- Now consider the relationship between a **product** and an **order**
  - An order could have more than one (**many**) products on it
  - A product could be on more than one (**many**) order
  - So the relationship between order and product is **many-to-many**
- **One-to-one** relationships also exist but are not very common in databases

What is an entity-relationship (E-R) diagram?

- An entity relationship diagram (**E-R**) is a **diagram that represents the entities** (tables) that will be in a database and the **relationships between these entities**
- The entities are drawn as **boxes** with the **entity** name in
- The relationships are drawn in as what is known as '**crow's feet notation**'

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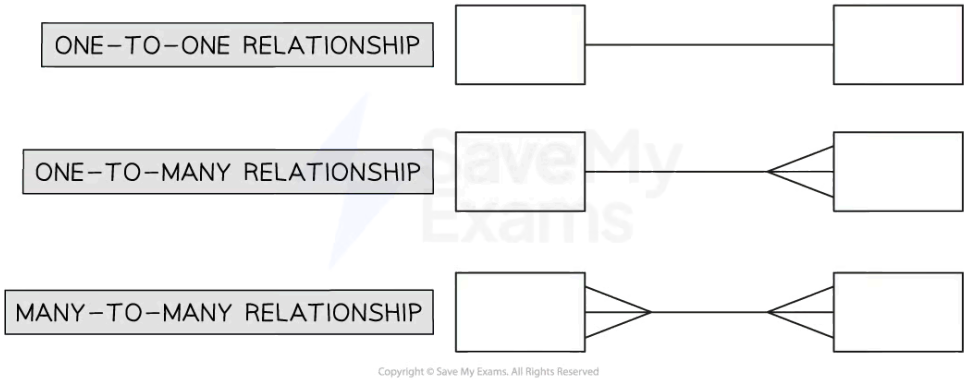
- This is how to draw the relationships in the exam:



- The **names** of the entities would go inside the boxes

Examiner Tips and Tricks

These diagrams are simple but tell us some important things about the database:

- The names of all the tables
- Which tables will have a foreign key - when an entity has a 'many' relationship against it that means it will have a foreign key in it that links to the primary key of the connected entity

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Normalised design

What is a normalised design?

- A **normalised design** is a database structure that:
  - **Organises data efficiently**
  - **Eliminates unnecessary duplication**
  - **Ensures data integrity**
  - **Supports easy updates and accurate relationships**
- You may be asked to produce a normalised design based on:
  - A **written description** of a system
  - A **table of unnormalised data**
  - A set of **existing flat tables**
- You need to be able to:
  - Identify **repeating groups** or **duplicated data**
  - Break down data into **separate related tables**
  - Assign appropriate **primary keys** and **foreign keys**
  - Show the database in **1NF, 2NF, and 3NF**
  - Clearly show **relationships** between the tables

Step-by-step process

Step	What to do	Why it matters
1	Read the scenario or table carefully	Understand what entities (real-world things) are involved
2	Identify the fields and any repeated/duplicated values	These usually indicate the need for more than one table
3	Apply **1NF** – remove repeating groups, separate fields	Each field should hold a single value
4	Apply **2NF** – remove partial dependencies	Make sure non-key fields depend on the whole primary key

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5	Apply **3NF** – remove transitive dependencies	Non-key fields should not depend on other non-key fields
6	Assign **primary keys** for each table	Every table needs a unique identifier
7	Use **foreign keys** to link related tables	Ensures relationships and referential integrity

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Example

- A student is enrolled on **multiple courses**
- Each course is taught by **a teacher**
- The table includes: StudentName, StudentID, CourseID, CourseName, TeacherName
- You would:
  - Identify **Entities**: Student, Course, Teacher
  - Create tables:
    - **Student** (StudentID, StudentName)
    - **Course** (CourseID, CourseName, TeacherID)
    - **Teacher** (TeacherID, TeacherName)
    - A link table: **StudentCourse** (StudentID, CourseID) – handles many-to-many

Examiner Tips and Tricks

- Always check for **repeating fields** or **duplicated values**
- Watch for **composite keys** in link tables (especially many-to-many relationships)
- Check that all fields in a table **depend only on the key, the whole key, and nothing but the key**
- Label each stage of your work (1NF → 2NF → 3NF)

First Normal Form (1NF)

What is first normal form?

- For a **table** to be in first normal form it must:
  - **Contain atomic values**
    - Each column in a table must contain single, indivisible values
  - **Have no repeating groups**
    - Columns must not contain arrays or lists of values

- **Have unique column names**
  - Each column must have a unique name within the table
- **Have a unique identifier (primary key)**
  - Each row must have a unique identifier to distinguish it from other rows
- This customers table below has **no primary key** and the name is stored in **one field** so is not atomic
- This table is **not** in first normal form

Table: Customers

name	phone	country
John Smith	07373 929122	UK
Iram Iravani	07234 543422	Iraq
Wu Zhang	04563 523427	China
Anne James	09378 482894	USA
Khalid Shirvani	02343 536522	France

- This customers table below **has a primary key** and the name is stored in **two fields** so it is **atomic**
- This table is in **first normal form**

Table: Customers

customer_id	forename	surname	phone	country
1	John	Smith	07373 929122	UK
2	Iram	Iravani	07234 543422	Iraq
3	Wu	Zhang	04563 523427	China
4	Anne	James	09378 482894	USA
5	Khalid	Shirvani	02343 536522	France

Second Normal Form (2NF)

What is second normal form?

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- For a table to be in second normal form it must:
  - **Fulfil all 1NF requirements**
  - **Only apply to tables with a compound primary key**
  - **Have full functional dependency**
    - **All non-prime attributes (attributes not part of the primary key) must be fully dependent on the primary key**
  - **Have no partial dependencies**
    - Non-prime attributes must not depend on only part of the primary key (in case of a composite primary key)
    - Separate tables should be created for partially dependent attributes
- In this table below, Course Title **only depends on part** of the compound primary key (the course code) and not the Date so this table **is not** in second normal form

Table: Course

Course	Date	Course Title	Room	Capacity	Available
SQL101	03/01/2020	SQL Basics	4A	12	4
DB202	03/01/2020	Database Design	7B	14	7
SQL101	04/05/2020	SQL Basics	7B	14	10
SQL101	15/05/2020	SQL Basics	12A	8	8
CS50	31/05/2020	C Programming	4A	12	11

- To turn this table into second normal form we will ensure:
  - Course Title is moved into its own **Course** table
  - A Session table is created to use the full key (Course, Date) for all time-specific info
  - No **partial dependencies** remain

Table: Course

Course	Course Title
SQL101	SQL Basics
DB202	Database Design

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CS50	C Programming

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Table: Session

Course	Date	Room	Capacity	Available
SQL101	03/01/2020	4A	12	4
DB202	03/01/2020	7B	14	7
SQL101	04/05/2020	7B	14	10
SQL101	15/05/2020	12A	8	8
CS50	31/05/2020	4A	12	11

Third Normal Form (3NF)

What is third normal form?

- For a table to be in third normal form it must:
  - **Fulfil all 2NF requirements**
  - **Have no transitive dependencies**
    - Non-prime attributes must not depend on other non-prime attributes
  - **Have each non-prime attribute dependent solely on the primary key, not on other non-prime attributes**
  - **Have separate tables for attributes with transitive dependencies, and the tables should be linked using a foreign key**
- In this table below, the certificate depends on the title - this a transitive dependency and so this table **is not** in third normal form

FilmID	Title	Certificate	Description
12034	Saw IV	18	Eighteen and over
12035	Spiderman 2	12A	Age 12 and over
12036	Shrek	U	Universal

- To turn this table into third normal form we will ensure:
  - Transitive dependency removed by **separating** Description into its own table

- The Film table only stores fields that **directly depend** on the key (FilmID)
- A **foreign key** (Certificate) is used to maintain the relationship

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Table: Film

FilmID	Title	Certificate
12034	Saw IV	18
12035	Spiderman 2	12A
12036	Shrek	U

Table: Certificate

Certificate	Description
18	Eighteen and over
12A	Age 12 and over
U	Universal

Worked Example

Match each normal form to its definition:

First Normal Form (1NF)	All fields are fully dependent on the primary key.
Second Normal Form (2NF)	There are no repeating groups of attributes.
Third Normal Form (3NF)	There are no partial dependencies.

[1]

Answer

First Normal Form (1NF)	There are no repeating groups of attributes.
Second Normal Form (2NF)	There are no partial dependencies.
Third Normal Form (3NF)	All fields are fully dependent on the primary key.

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